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Papers 262 (conclusion), 263, 264, 265, 266 (opening)

FOR THE QUINTIC EQUATION $(a, b, c, d, e, f \mid v, 1)^5 = 0$, *the equation of differences is,*

θ^{10}	θ^{9}	θ^{8}	θ^{7}	θ^{6}	θ^{5}	θ^{4}
$a^8\times$	$100a^6\times$	$50a^4\times$	$2500a^2\times$	$125\times$	$625\times$	$2500\times$
$1 + 1$	$ac + 1$ $b^2 - 1$	$a^3e + 1$ $a^2bd - 4$ $a^2c^2 + 78$ $ab^2c - 150$ $b^4 + 75$	$a^4ce + 1$ $a^4d^2 + 1$ $a^3b^2e - 1$ $a^3bcd - 10$ $a^3c^3 + 32$ $a^2b^3d + 8$ $a^2b^2c^2 - 81$ $ab^4c + 75$ $b^6 - 25$	$a^6df + 16$ $a^6e^2 - 19$ $a^5bcf - 48$ $a^5bde + 72$ $a^5c^2e + 496$ $a^5cd^2 + 736$ $a^4b^3f + 32$ $a^4b^2ce - 980$ $a^4b^2d^2 - 880$ $a^4bc^2d - 6400$ $a^4c^4 + 7600$ $a^3b^4e + 450$ $a^3b^3cd + 11600$ $a^3b^2c^3 - 20800$ $a^2b^5d - 5000$ $a^2b^4c^2 + 22500$ $ab^6c - 12500$ $b^8 + 3125$	$a^6f^2 + 1$ $a^5bef - 10$ $a^5cdf + 64$ $a^5ce^2 - 144$ $a^5d^2e + 208$ $a^4b^2df - 44$ $a^4b^2e^2 + 169$ $a^4bc^2f - 192$ $a^4bcde - 416$ $a^4bd^3 - 832$ $a^4c^3e + 1568$ $a^4c^2d^2 + 1888$ $a^3b^3ef + 300$ $a^3b^3de - 100$ $a^3b^2c^2e - 3600$ $a^3b^2cd^2 + 800$ $a^3bc^3d - 17600$ $a^3c^5 + 10560$ $a^2b^5f - 120$ $a^2b^4ce + 3300$ $a^2b^4d^2 - 300$ $a^2b^3c^2d + 38400$ $a^2b^2c^4 - 26400$ $ab^6e - 1000$ $ab^5cd - 33000$ $ab^4c^3 + 24000$ $b^7d + 10000$ $b^6c^2 - 7500$	$a^5cf^2 + 7$ $a^5def - 19$ $a^5e^3 + 2$ $a^4b^2f^2 - 7$ $a^4bcef - 13$ $a^4bd^2f + 76$ $a^4bde^2 + 71$ $a^4c^2df + 96$ $a^4c^2e^2 - 522$ $a^4cd^2e + 1416$ $a^4d^4 - 212$ $a^3b^3ef + 32$ $a^3b^2ce^2 + 970$ $a^3b^2cdf - 394$ $a^3b^2d^2e - 1890$ $a^3bc^3f - 288$ $a^3bc^2de - 4800$ $a^3bcd^3 - 3120$ $a^3c^4e + 3880$ $a^3c^3d^2 + 2080$ $a^2b^4df + 165$ $a^2b^4e^2 - 525$ $a^2b^3c^2f + 970$ $a^2b^3cde + 8250$ $a^2b^3d^3 + 4600$ $a^2b^2c^3e - 7600$ $a^2b^2c^2d^2 + 17400$ $a^2bc^4d - 28000$ $a^2c^6 + 10000$ $ab^5cf - 875$ $ab^5de - 2625$ $ab^4c^2e + 4000$ $ab^4cd^2 - 36000$ $ab^3c^3d + 50000$ $ab^2c^5 - 18000$ $b^7f + 250$ $b^6ce - 625$ $b^6d^2 + 13750$ $b^5c^2d - 20000$ $b^4c^4 + 7500$
± 1	± 1	± 154	± 117	± 46627	± 91258	± 125515

$0 = ((\theta, 1))^{10} = 0$, viz. the function in θ is

θ^3	θ^2	θ	θ^0
$6250\times$	$62500\times$	$62500\times$	$3125\times$
$a^5ef^2 - 3$	$a^4cef^2 - 4$	$a^4df^3 - 1$	$a^4f^4 + 1$
$a^4bdf^2 + 12$	$a^4d^2f^2 + 3$	$a^4e^2f^2 + 2$	$a^3bef^3 - 20$
$a^4be^2f + 30$	$a^4de^2f + 4$	$a^3bcf^3 + 3$	$a^3cdf^3 - 120$
$a^4c^2f^2 + 40$	$a^4e^4 + 4$	$a^3bdef^2 - 1$	$a^3ce^2f^2 + 160$
$a^4cdef - 172$	$a^3b^2ef^2 + 4$	$a^3be^3f - 20$	$a^3d^2ef^2 + 360$
$a^4ce^3 - 16$	$a^3bcdf^2 - 2$	$a^3c^2ef^2 - 30$	$a^3de^3f - 640$
$a^4d^3f + 112$	$a^3bce^2f + 28$	$a^3cd^2f^2 + 48$	$a^3e^5 + 256$
$a^4d^2e^2 + 228$	$a^3bd^2ef - 62$	$a^3cde^2f + 8$	$a^2b^2df^3 + 160$
$a^3b^2cf^2 - 98$	$a^3bde^3 - 84$	$a^3ce^4 + 32$	$a^2b^2e^2f^2 - 10$
$a^3b^2def - 8$	$a^3c^3f^2 + 28$	$a^3d^3ef - 36$	$a^2bc^2f^3 + 360$
$a^3b^2e^3 - 59$	$a^3c^2def - 132$	$a^3d^2e^3 - 8$	$a^2bcdef^2 - 1640$
$a^3bc^2ef + 116$	$a^3c^2e^3 - 72$	$a^2d^3f^3 - 2$	$a^2bce^3f + 320$
$a^3bcd^2f - 320$	$a^3cd^3f + 96$	$a^2b^2cef^2 + 39$	$a^2bd^3f^2 - 1440$
$a^3bcde^2 - 316$	$a^3cd^2e^2 + 384$	$a^2b^2d^2f^2 - 46$	$a^2bd^2e^2f + 4080$
$a^3bd^3e - 2384$	$a^3d^4e - 216$	$a^2b^2de^2f + 117$	$a^2bde^4 - 1920$
$a^3c^3df + 128$	$a^2b^3df^2 - 4$	$a^2b^2e^4 + 18$	$a^2c^3ef^2 - 1440$
$a^3c^3e^2 - 1232$	$a^2b^3e^2f - 32$	$a^2bc^2df^2 - 168$	$a^2c^2d^2f^2 + 2640$
$a^3c^2d^2e + 4896$	$a^2b^2c^2f^2 - 81$	$a^2bc^2e^2f + 276$	$a^2c^2de^2f + 4480$
$a^3cd^4 - 1632$	$a^2b^2cdef + 348$	$a^2bcd^2ef - 220$	$a^2c^2e^4 - 2560$
$a^2b^4f^2 + 49$	$a^2b^2ce^3 + 200$	$a^2bcde^3 - 504$	$a^2cd^3ef - 10080$
$a^2b^3cef + 180$	$a^2b^2d^3f + 28$	$a^2bd^4f + 144$	$a^2cd^2e^3 + 5760$
$a^2b^3d^2f + 224$	$a^2b^2d^2e^2 + 275$	$a^2bd^3e^2 + 276$	$a^2d^5f + 3456$
$a^2b^3de^2 + 460$	$a^2bc^3ef + 116$	$a^2c^4f^2 + 108$	$a^2d^4e^2 - 2160$
$a^2b^2c^2df + 344$	$a^2bc^2d^2f - 336$	$a^2c^3def - 264$	$ab^3cf^3 - 640$
$a^2b^2c^2e^2 + 3880$	$a^2bc^2de^2 - 780$	$a^2c^3e^3 + 448$	$ab^3def^2 + 320$
$a^2b^2cd^2e + 3000$	$a^2bcd^3e - 960$	$a^2c^2d^3f + 96$	$ab^3e^3f - 180$
$a^2b^2d^4 + 6400$	$a^2bd^5 + 864$	$a^2c^2d^2e^2 + 1736$	$ab^2c^2ef^2 + 4080$
$a^2bc^4f - 384$	$a^2c^4df + 64$	$a^2cd^4e - 1584$	$ab^2cd^2f^2 + 4480$
$a^2bc^3de - 20160$	$a^2c^4e^2 - 60$	$a^2d^6 + 432$	$ab^2cde^2f - 14920$
$a^2c^5e + 8480$	$a^2c^3d^2e + 1120$	$ab^4ef^2 - 12$	$ab^2ce^4 + 7200$
$a^2c^4d^2 - 1120$	$a^2c^2d^4 - 720$	$ab^3cdf^2 + 264$	$ab^2d^3ef + 960$
$ab^5ef - 170$	$ab^4cf^2 + 84$	$ab^3ce^2f - 615$	$ab^2d^2e^3 - 600$
$ab^4cdf - 860$	$ab^4def - 100$	$ab^3d^2ef - 12$	$abc^3df^2 - 10080$
$ab^4ce^2 - 4450$	$ab^4e^3 - 60$	$ab^3de^3 + 45$	$abc^3e^2f + 960$
$ab^4d^2e - 2700$	$ab^3c^2ef - 310$	$ab^2c^3f^2 - 180$	$abc^2d^2ef + 28480$
$ab^3c^2de + 23400$	$ab^3cd^2f - 100$	$ab^2c^2def + 1188$	$abc^2de^3 - 16000$
$ab^3c^3f + 680$	$ab^3cde^2 - 890$	$ab^2c^2e^3 + 1410$	$abcd^4f - 11520$
$ab^3cd^3 - 29600$	$ab^3d^4e - 140$	$ab^2cd^3f - 616$	$abcd^3e^2 + 7200$
$ab^2c^4e - 11200$	$ab^2c^3df + 520$	$ab^2cd^2e^2 - 1140$	$ac^5f^2 + 3456$
$ab^2c^3d^2 + 44800$	$ab^2c^3e^2 + 1120$	$ab^2d^4e + 120$	$ac^4def - 11520$
$abc^5d - 27200$	$ab^2c^2d^2e + 4920$	$abc^4ef - 288$	$ac^4e^3 + 6400$
$ac^7 + 6400$	$ab^2cd^4 - 3120$	$abc^3d^2f + 192$	$ac^3d^3f + 5120$
$b^6df + 400$	$abc^5f - 192$	$abc^3de^2 - 3720$	$ac^3d^2e^2 - 3200$
$b^6e^2 + 1625$	$abc^4de - 6560$	$abc^2d^3e + 4640$	$b^5f^3 + 256$
$b^5c^2f - 300$	$abc^3d^3 + 4160$	$abcd^5 - 1440$	$b^4cef^2 - 1920$
$b^5cde - 7500$	$ac^6e + 1920$	$ac^5e^2 + 1440$	$b^4d^2f^2 - 2560$
$b^5d^3 + 14000$	$ac^5d^2 - 1280$	$ac^4d^2e - 1920$	$b^4de^2f + 7200$
$b^4c^3e + 4000$	$b^6f^2 - 28$	$ac^3d^4 + 640$	$b^4e^4 - 3375$
$b^4c^2d^2 - 24000$	$b^5cef + 140$	$b^5df^3 - 96$	$b^3c^2df^2 + 5760$
$b^3c^4d + 16000$	$b^5d^2f + 120$	$b^5e^2f + 270$	$b^3c^2e^2f - 600$
$b^2c^6 - 4000$	$b^5de^2 + 600$	$b^4c^2f^2 + 72$	$b^3cd^2ef - 16000$
	$b^4c^2df - 320$	$b^4cdef - 600$	$b^3cde^3 + 9000$
	$b^4c^2e^2 - 625$	$b^4ce^3 - 675$	$b^3d^4f + 6400$
	$b^4cd^2e - 2700$	$b^4d^3f + 320$	$b^3d^3e^2 - 4000$
	$b^4d^4 + 1700$	$b^4d^2e^2 + 450$	$b^2c^4f^2 - 2160$
	$b^3c^4f + 120$	$b^3c^3ef + 180$	$b^2c^3def + 7200$
	$b^3c^3de + 3800$	$b^3c^2d^2f - 120$	$b^2c^3e^3 - 4000$
	$b^3c^2d^3 - 2400$	$b^3c^2de^2 + 2100$	$b^2c^2d^3f - 3200$
	$b^2c^5e - 1200$	$b^3cd^3e - 2600$	$b^2c^2d^2e^2 + 2000$
	$b^2c^4d^2 + 800$	$b^3d^5 + 800$	
		$b^2c^4e^2 - 900$	
		$b^2c^3d^2e + 1200$	
		$b^2c^2d^4 - 400$	
± 139884	± 23570	± 18666	± 128505

$(\theta, 1)^{10}$.

[where in each column after the first the sum of the positive coefficients is equal to that of the negative coefficients, and I have inserted at the foot this sum with the sign $+$. A single coefficient -7500 in place of -3750 has been corrected.]

The coefficients in the preceding equations of differences are functions of the seminvariants of the quantics to which they belong; for instance, in the case of the quartic, the coefficient of θ^4 is

$$8\,a^2\big[\,a^2(ae-4bd+3c^2)+96\,(ac-b^2)\big],$$

that of θ^3 is

$$32\big[-13\,a^2(ace-ad^2-b^2e+2bcd-c^3)+16\,a^2(ac-b^2)(ae-4bd+3c^2)+128\,(ac-b^2)^3\big],$$

and so for the other coefficients; and by replacing each seminvariant by the covariant to which it belongs, we pass from the solution of the original problem of finding the equation for $\theta=(\alpha-\beta)^2$, to that of the problem of finding the equation for

$$\theta=\frac{(x-\alpha y)^2(x-\beta y)^2}{(x-\gamma y)^2\cdots}.$$

The results are as follows: —

For the quadric $(a,b,c\mid x,y)^2$, the equation in θ is, $0=$

$$\big(U^2,\ 4\square\big)(\theta,\ 1)^1,$$

where U is the quadric, $\square$ the discriminant.

For the cubic $(a,b,c,d\mid x,y)^3$, the equation in θ is, $0=$

$$\big(U^4,\ 18\,U^2H,\ 81\,H^2,\ 27\square U^2\big)(\theta,\ 1)^3,$$

where U is the cubic, H the Hessian, $\square$ the discriminant.

For the quartic $(a,b,c,d,e\mid x,y)^4$, the equation in θ is, $0=$

$$\left\{\begin{array}{l} U^7, \\ 48\,U^5H, \\ 32\big(-13\,U^4J+16\,U^3HI+128\,H^3\big), \\ 16\big(-7\,U^3I^2-288\,UHJ+384\,H^2I\big), \\ 1152\big(-3\,UIJ+2\,HI^2\big), \\ 256\,\big(I^3-27\,J^2\big) \end{array}\right\}(\theta,\ 1)^4,$$

where U is the quartic, H the Hessian, I and J the quadrinvariant and the cubinvariant respectively.

For the quintic $(a, b, c, d, e, f \mid x, y)^5$, the equation in θ, as far as it can be expressed in terms of known covariants, is, $0 =$

$$\left\{\begin{array}{l} U^8, \\ 100\, U^6\,(\text{Tab. No. } 15), \\ 50\, U^4\big[U^2\,(\text{Tab. No. } 14) + 75\,(\text{Tab. No. } 15)^2\big], \\ \quad\vdots \\ 3125 \text{ Discriminant } (= \text{Tab. No. } 26) \end{array}\right\} (\theta,\ 1)^{10},$$

where the Tables referred to are those in my Second Memoir on Quantics, [141]; for the completed equation see post p. 261].

The form of the preceding results may be modified by writing $\theta = \frac{1}{a^2}\, U^2 \vartheta$; we have thus the equations for

$$\vartheta = a^2\,(\alpha-\beta)^2\,(x-\gamma y)^2 \cdots;$$

thus for example, in the case of the cubic $(a, b, c, d \mid x, y)^3$, the equation for ϑ $[= a^2(\alpha-\beta)^2(x-\gamma y)^2]$ is

$$0 = \big(1,\ 18H,\ 81H^2,\ 27\,\square\, U^{-2}\big)(\vartheta,\ 1)^3;$$

this equation may be written

$$(\vartheta + 9H)^3 + 27\,\square\, U^{-2} = 0;$$

or putting $v = U^{2/3}\vartheta$, we have

$$v^3 + 9Hv + U^2\sqrt{-27\,\square} = 0,$$

an equation the roots of which are

$$a\,(\alpha-\beta)(x-\gamma y), \quad a\,(\beta-\gamma)(x-\alpha y), \quad a\,(\gamma-\alpha)(x-\beta y),$$

and which leads to the formula, given in my fifth Memoir on Quantics, [156], for the solution of a cubic equation. But this decomposition of the equation in ϑ is peculiar to the cubic.

The equation of differences for an equation of any order may be found by the following entirely distinct method. Let $\varphi v = (* \mid v, 1)^n = 0$ be for shortness represented by $\varphi v = 0$, and let x, y be any two distinct roots; we have not only $\varphi x = 0$, $\varphi y = 0$, but also $\varphi x + \varphi y = 0$, $\dfrac{\varphi x - \varphi y}{x - y} = 0$. Writing $\theta = (x-y)^2$, $s = x + y$, we have

$$x = \tfrac{1}{2}(s + \sqrt{\theta}), \qquad y = \tfrac{1}{2}(s - \sqrt{\theta}),$$

values which are to be substituted for x, y in the equations

$$\varphi x + \varphi y = 0, \qquad \frac{\varphi x - \varphi y}{x - y} = 0.$$

We have thus two equations rational in s and θ, and the elimination between them of the quantity s leads to the required equation in θ. But it is proper to modify the form of the system; in fact the two equations are, as regards s, the first of them of the degree n, the second of the degree $n-1$; but if we write

$$\varphi x + \varphi y = 0, \qquad \frac{\varphi x - \varphi y}{x - y} = 0,$$

then each of the equations will be of the same degree $n-1$ in s.

For instance, let $\varphi v = (a, b, c, d \mid v, 1)^3$; then $x = \frac{1}{2}(s + \sqrt{\theta})$, $y = \frac{1}{2}(s - \sqrt{\theta})$; the equations $\varphi x + \varphi y = 0$, $\dfrac{\varphi x - \varphi y}{x - y} = 0$, are

$$s^3.a + 3s^2.2b + 3s\,(4c + a\theta) + 8d + 6b\theta = 0,$$
$$3s^2.a + 3s.4b + 12c + a\theta = 0;$$

and multiplying the first equation by 3 and the second by $-s$, adding and dividing by 2, we have an equation

$$s^2.3b + s\,(12c + 4a\theta) + 12d + 9b\theta = 0.$$

The second equation and this equation may be written

$$(3a,\ 12b,\ 12c + a\theta \mid s, 1)^2 = 0,$$
$$(3b,\ 12c + 4a\theta,\ 12d + 9b\theta \mid s, 1)^2 = 0,$$

and the elimination of s from these equations gives the required equation in θ. The result may be obtained under either of the two forms,

$$\{a^2\theta^2 + (15ac - 27b^2)\,\theta - 36\,(bd - c^2)\}\{a^2\theta + 3\,(ac - b^2)\} + 3\{2ab\theta + 3\,(ad - bc)\}^2 = 0$$

and

$$\{4a^2\theta + (24ac - 27b^2)\,\theta - 36\,(bd - c^2)\}\left[a^2\theta + 12\,(ac - b^2)\right] + 3\{ab\theta + 6\,(ad - bc)\}^2 = 0,$$

the expansions of which respectively coincide with the before-mentioned result.

In the case of the quartic equation $\varphi v = (a, b, c, d, e \mid v, 1)^4 = 0$, we have

$$s^4.a + 4s^3.2b + 6s^2\,(4c + a\theta) + 4s\,(8d + 6b\theta) + 16e + 24c\theta + a\theta^2 = 0,$$
$$4s^3.a + 6s^2.4b + 4s\,(12c + a\theta) + 32d + 8b\theta = 0,$$

from which we derive another cubic equation; the two cubic equations thus are

$$(4a,\ 24b,\ 48c + 4a\theta,\ 32d + 8b\theta \mid s, 1)^3 = 0,$$
$$(4b,\ 24c + 10a\theta,\ 48d + 44b\theta,\ 32e + 48c\theta + 2a\theta^2 \mid s, 1)^3 = 0,$$

from which, if s be eliminated, we have the equation in θ.

Similarly, for the quintic equation $\varphi v = (a, b, c, d, e, f \mid v, 1)^5 = 0$, we have

$$s^5.a + 5s^4.2b + 10s^3(4c + a\theta) + 10s^2(8d + 6b\theta) \\ + 5s(16e + 24c\theta + a\theta^2) + 32f + 80d\theta + 10b\theta^2 = 0,$$

$$5s^4.a + 10s^3.4b + 10s^2(12c + a\theta) + 5s(32d + 8b\theta) + 80e + 40c\theta + a\theta^2 = 0,$$

from which we derive another quartic equation; the two equations are

$$(5a,\ 40b,\ 120c + 10a\theta,\ 160d + 40b\theta,\ 80e + 40c\theta + a\theta^2 \mid s, 1)^4 = 0,$$

$$(5b,\ 40c + 20a\theta,\ 120d + 110b\theta,\ 160e + 280c\theta + 12a\theta^2,\ 80f + 200d\theta + 25b\theta^2 \mid s, 1)^4 = 0,$$

from which, if s be eliminated, we have the equation in θ.

But we may apply Bézout's method to the two equations each of the order $n-1$, which results from the equation of the order n, in the present case the equation $\varphi v = (* \mid v, 1)^n = 0$. The process is as follows: —Suppose, in general, that s is to be eliminated from the two equations

$$Fs = 0, \qquad Gs = 0,$$

each of the order $n-1$; it is only necessary to form the expression

$$\frac{Fs\,Gs' - Fs'\,Gs}{s - s'}$$

which will be a function of s, s' of the form

$$(a_{1,1},\ a_{1,2},\ \ldots,\ a_{n-1,n-1} \mid s, 1)^{n-2}(s', 1)^{n-2}$$

where the coefficients are such that $a_{i,j} = a_{j,i}$; and by equating to zero the determinant formed with these coefficients, we have the result of the elimination.

In the present case, writing for a moment $\frac{1}{2}(s + \sqrt{\theta}) = A$, $\frac{1}{2}(s - \sqrt{\theta}) = B$, and in like manner $\frac{1}{2}(s' + \sqrt{\theta}) = A'$, $\frac{1}{2}(s' - \sqrt{\theta}) = B'$, we have

$$Fs = \varphi(A + B) - \frac{\varphi(A - B)}{\sqrt{\theta}}, \qquad Gs = \frac{A - B}{\sqrt{\theta}},$$

$$Fs' = \varphi(A' + B') - \frac{\varphi(A' - B')}{\sqrt{\theta}}, \qquad Gs' = \frac{A' - B'}{\sqrt{\theta}},$$

and therefore

$$Fs\,Gs' - Fs'\,Gs = \frac{(A + B)(A' - B') - (A' + B')(A - B)}{\sqrt{\theta}} \cdots,$$

or reducing and dividing by $s - s'$,

$$\frac{Fs\,Gs' - Fs'\,Gs}{s - s'} = \frac{1}{\sqrt{\theta}(s - s')} \cdot \frac{AB' - A'B}{1} \cdot \frac{(A - B)(A' - B')}{\theta} \cdots.$$

Hence, substituting for A, B, A', B' these values, we have the expression

$$\frac{2m}{\sqrt{\theta}} \cdot \frac{\cdots}{\theta}$$

which is of the form

$$(a_{1,1},\ \ldots,\ a_{n-1,n-1} \mid s, 1)^{n-2}\,(s', 1)^{n-2},$$

and equating to zero the determinant formed with the coefficients, we have an equation in θ which is the equation of differences of the given equation $\varphi v = 0$. For instance, if the given equation is $\varphi v = (a, b, c, d \mid v, 1)^3 = 0$, then we have

$$\begin{aligned}
\varphi\big(\tfrac{1}{2}(s+\sqrt{\theta})\big) &= \big(a,\ 2b + a\sqrt{\theta},\ 4c + 4b\sqrt{\theta} + a\theta,\ 8d + 12c\sqrt{\theta} + 6b\theta + a\theta\sqrt{\theta}\ \mid s, 1\big)^3 \\
&= (A,\ B,\ C,\ D \mid s, 1)^3, \\
\varphi\big(\tfrac{1}{2}(s-\sqrt{\theta})\big) &= \big(a,\ 2b - a\sqrt{\theta},\ 4c - 4b\sqrt{\theta} + a\theta,\ 8d - 12c\sqrt{\theta} + 6b\theta - a\theta\sqrt{\theta}\ \mid s, 1\big)^3 \\
&= (A',\ B',\ C',\ D' \mid s, 1)^3;
\end{aligned}$$

and the function in s, s' is

$$\begin{aligned}
&3\,(AB' - A'B)\,ss'^2 \\
&+ 3\,(AC' - A'C)\,ss'\,(s + s') \\
&+ \big[(AD' - A'D) + 9\,(BC' - B'C)\big]\big(s^2 + ss' + s'^2\big) \\
&+ 3\,(BD' - B'D)\,(s + s') \\
&+ 3\,(CD' - C'D) \\
&+ \tfrac{1}{\theta}\,(A - A',\ B - B',\ C - C',\ D - D' \mid s, 1)^3\,(A - A',\ B - B',\ C - C',\ D - D' \mid s', 1)^3,
\end{aligned}$$

which is equal to

$$\begin{aligned}
&- 12\,ab\,ss'\,(s + s') \\
&+ 12\,(-12ac - a^2\theta)\,(s^2 + ss' + s'^2) \\
&+ (36ac - 72b^2 + 9a^2\theta)\,ss' \\
&+ (24ad - 72bc + 12ab\theta)\,(s + s') \\
&+ 96bd - 144c^2 + (-48ac + 72b^2)\,\theta - 3a^2\theta^2 \\
&+ 4\,(3as^2 + 12bs + 12c + a\theta)\,(3as'^2 + 12bs' + 12c + a\theta);
\end{aligned}$$

or reducing and dividing by 32, this is

$$\left\{9\,(ac-b^2)+3a^2\theta\right\}ss'+\left\{9\,(ad-bc)+6ab\theta\right\}(s+s')+36\,(bd-c^2)+(-15ac+27b^2)\,\theta-a^2\theta^2,$$

the terms in s^2, s'^2 disappearing, as they should do. Writing this under the form

$$\begin{pmatrix} 9\,(ac-b^2)+3a^2\theta, & 9\,(ad-bc)+6ab\theta \\ 9\,(ad-bc)+6ab\theta, & 36\,(bd-c^2)+(-15ac+27b^2)\,\theta-a^2\theta^2 \end{pmatrix}(s,1)\,(s',1)$$

and equating the determinant to zero, we have the required equation in θ: the form is that obtained by the ordinary process of applying Bézout's method to the two equations $(3a, 12b, 12c+a\theta \mid s,1)^2=0$, $(3b, 12c+4a\theta, 12d+9b\theta \mid s,1)^2=0$, being in fact the before-mentioned equation

$$\left\{a^2\theta^2+(15ac-27b^2)\,\theta-36\,(bd-c^2)\right\}\left\{a^2\theta+3\,(ac-b^2)\right\}+3\left\{2ab\theta+3\,(ad-bc)\right\}^2=0.$$

But, as already remarked, this elimination process is less convenient for the complete development of the result, than the method first explained in the present memoir.

[The equation p. 257, changing the notation and inserting the omitted coefficients, becomes $0=$

$\theta^{10}\cdot 100\times\ (A^2B+1,\ A^2C+1)$;

$\theta^{9}\cdot 50\times\ (A^2B+1,\ A^2CD+75)$;

$\theta^{7}\cdot 2500\times\ (A^2BC+1,\ A^2F^2+1,\ A^2CD+29)$;

$\theta^{6}\cdot 125\times\ (A^2BI+1169,\ A^2BC^2+450,\ CF^2+800,\ C^4+6323)$;

$\theta^{5}\cdot 625\times\ (A^2G+1601,\ AFHD+60,\ BF^2+112480,\ BACD+3620)$;

$\theta^{4}\cdot 12500\times\ (A^2BH-1545,\ A^2J,\ AECI+230,\ AECJ+275,\ BEF+1225,$
$\quad +11650,\ CAHB^2+1125)$;

$\theta^{3}\cdot 6250\times\ (A^2BMG-33267,\ -21450200,\ AEBGCJD+23820000,\ BCH+1625,\ C^2G)$;

$\theta^{2}\cdot 62500\times\ (ABDJG-20,\ +24,\ BCG+19,\ B^2H-36)$;

$\theta^{1}\cdot 62500\times\ (B^2JG-162,\ GH,\ DJ,\ CM-28)$;

$\theta^{0}\cdot 3125\times\ (Q)$,

where the capital letters denote the covariants of the quantic as explained 141 and 143.]

Note. The exact column layout of the printed page 261 is here transposed into a single sequential listing for clarity. The detailed monomial structure and signs follow the original.

263.

Demonstration of a Theorem in Finite Differences.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CL. (for the year 1860), pp. 321–323: printed as a note to Sir J. W. F. Herschel's Memoir "On the Formulae investigated by Dr Brinkley for the general Term in the Development of Lagrange's Expression for the Summation of Series and for Successive Integrations," pp. 319–321.]

The formula (B) [of Sir J. W. F. Herschel's Memoir], substituting therein for Δx the value $([x])\Delta\nabla^x 0^n$, becomes

$$\frac{1}{[n-1]}\left\{\frac{1}{[x-1]}[n+1]\nabla^x - \frac{1}{[x-2]}[n+2]\nabla^x \text{ \&c.}\right\}0^n,$$

or, as this may be written,

$$\frac{1}{[n-1]}\left\{\frac{[x]}{1\cdot 2}\nabla^x 0^n - \frac{[x]}{2\cdot 3}\nabla^x 0^n + \cdots\right\},$$

or, inserting a first term which vanishes except in the case $x = 0$, and which is required in order that the formula may hold good for this particular value,

$$\frac{1}{[n-1]}\left\{[0][x]\,0^n - [1][x-1]\,(n+1)^x + [2][x-2]\,(n+2)^x - \text{\&c.}\right\},$$

where the series on the right-hand side need only be continued up to the term containing $\nabla^x 0^n$, since the subsequent terms vanish. [In these formulæ and throughout the present paper $[x]$ is written to denote the factorial $[x]^x$ or $\Pi(x)$, and so in other cases.]

Now $\nabla^x 0^n$, or $(\frac{e^t-1}{e^t})^x 0^n$, is equal to $[x]\times$ coef. t^n in $(\frac{e^t-1}{e^t})^x$, and so $\nabla^x 0^n$, or $(\frac{e^t-1}{e^t})^x 0^n$, is equal to $[x]\times$ coef. t^n in $(\frac{e^t-1}{e^t})^x$.

Hence, putting $R = \dfrac{e^t - 1}{t}$, the last-mentioned formula will be true if, as regards the term which contains t^n, we have

$$R^{-n} = \frac{1}{[n-1]}\left\{\frac{1}{[0][x]}n - \frac{1}{[1][x-1]}(n+1)\,R + \frac{1}{[2][x-2]}(n+2)\,R^2 - \&\text{c.}\right\},$$

the series on the right-hand side being continued up to the term in R^x. This formula is, in fact, true if R, instead of being restricted to denote $\dfrac{e^t-1}{t}$, denotes any function whatever of the form $1 + bt + ct^2 + \&\text{c.}$, and it is true not only for the term in t^n, but for all the powers of t not higher than t^x. And, moreover, R^{-n} may denote any positive or negative integral or fractional power of R. In fact, the formula (assuming for a moment the truth of it) shows that the expansion of any power whatever of a series of the form in question, can be obtained by means of the expansions of the successive positive integer powers of the same series: the existence of such a formula (at least for negative powers) was indicated by Eisenstein, *Crelle*, t. XXXIX. p. 181 (1850), and the formula itself, in a slightly different form, was obtained in a very simple manner by Professor Sylvester in his paper, "Development of an idea of Eisenstein," *Quart. Math. Journ.* t. I. p. 199 (1855); the demonstration was in fact as follows, viz. writing

$$R^n = \big(1 + (R-1)\big)^n = 1 + n\,(R-1) + \frac{n\,(n-1)}{1\cdot 2}\,(R-1)^2 + \&\text{c.}:$$

if we attend only to the terms involving powers of t not higher than t^x, the series on the right-hand side needs only to be continued up to the term involving $(R-1)^x$, and the right side being thus converted into a rational and integral function of R, it may be developed in a series of powers of R (the highest power being of course R^x), and the coefficients of the several powers are finite series which admit of summation; this gives the required formula. But there is an easier method; the process shows that the series on the right-hand side, continued as above up to the term involving t^x, is, as regards n, a rational and integral function of the degree x; and by Lagrange's interpolation formula, any rational and integral function of n of the degree x, can be at once expressed in terms of the values corresponding to $x+1$ particular values of n. The investigation will be as follows: — Let R denote a series of the form $1 + bt + ct^2 + \&\text{c.}$, and let R^n denote the development of the nth power of R, continued up to the term containing t^x, the terms involving higher powers of t being rejected: R^0, R^1, R^2, &c., and generally R^s, will in like manner denote the developments of these powers up to the term involving t^x, or what is the same thing, they will be the values of R^n, corresponding to $n = 0, 1, 2, \ldots x$. By what precedes R^n is a rational and integral function of n of the degree x, and it can therefore be expressed in terms of the values $R^0, R^1, R^2, \ldots R^x$, which correspond to $n = 0, 1, 2, \ldots x$. Let s have any one of the last-mentioned values, then the expression

$$\frac{n\,.\,n-1\,.\,n-2\cdots n-x}{n-s}\cdot\frac{1}{s\,.\,s-1\cdots 2\,.\,1\,.\,-1\,.\,-2\cdots-(x-s)}\,R^s,$$

which, as regards n, is a rational and integral function of the degree x (the factor

$n-s$, which occurs in the numerator and in the denominator being of course omitted), vanishes for each of the values $n = 0, 1, 2, \dots x$, except only for the value $n = s$, in which case it becomes equal to unity. The required formula is thus seen to be

$$R^n = \sum \left\{ \frac{n\,.\,n-1\,.\,n-2\,\cdots\,n-x}{n-s} \cdot \frac{1}{s\,.\,s-1\,\cdots\,2\,.\,1\,.\,-1\,.\,-2\,\cdots\,-(x-s)}\, R^s \right\},$$

where the summation extends to the several values $s = 0, 1, 2, \dots x$; or, what is the same thing, it is

$$R^n = \sum \left\{ \frac{n\,.\,n-1\,.\,n-2\,\cdots\,n-x}{n-s} \cdot \frac{(-)^{x-s}}{1\cdot 2\cdots s\,.\,1\cdot 2\cdots (x-s)}\, R^s \right\},$$

or, as this may be written,

$$R^n = \sum \left\{ \frac{n\,.\,n-1\,.\,n-2\,\cdots\,n-x}{n+s} \cdot \frac{(-)^{x-s}}{1\cdot 2\cdots s\,.\,1\cdot 2\cdots (x-s)}\, R^s \right\},$$

or substituting for s the values $0, 1, 2, \dots x$, the formula is

$$R^n = \frac{n\,.\,n-1\,.\,n-2\,\cdots\,n-x}{[n-1]} \left\{ \frac{1}{[0][x]\,n} - \frac{1}{[1][x-1]\,(n+1)} R + \frac{1}{[2][x-2]\,(n+2)} R^2 - \cdots \right\},$$

continued up to the term involving R^x, which is the theorem in question.

264.

On an Extension of Arbogast's Method of Derivations.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. (for the year 1861), pp. 37–43, Received October 18,—Read December 13, 1860.]

Arbogast's Method of Derivations was devised by him with a view to the development of a function $\varphi(a+bx+cx^2+\cdots)$, but it is at least as useful for the formation of only the literal parts of the coefficients, or, what is the same thing, the combinations of a given degree and weight in the letters $(a, b, c, d, \ldots)$, the weights of the successive letters being $0, 1, 2, 3$, &c. Thus instead of applying the method to finding the coefficients

$$a^4, \quad 4a^3b, \quad 4a^3c+6a^2b^2, \quad \text{\&c.},$$

we may apply it merely to finding the sets of terms

$$a^4, \quad a^3b, \quad a^3c, \quad a^2b^2, \quad \text{\&c.}$$

To derive any column from the one which immediately precedes it, we operate on a letter by changing it into its immediate successor in the alphabet, and we must in each term operate on the last letter, and also, when the last but one letter in the term is the immediate antecessor in the alphabet of the last letter (but in this case only), operate on the last but one letter. Thus a^3c gives a^3d, but a^2b^2 gives a^2bc and a^3c, and the next succeeding column is therefore

$$a^3d, \qquad a^2bc.$$

If the series of letters is finite, and the last letter of the term is also the last letter of the series, then it is impossible to operate on the last letter of the term, but the last but one letter (when the foregoing rule applies to it) is still to be operated on; and if the rule does not apply, then the term does not give rise to a term in the succeeding column; the operations will at length terminate, and a complete series of columns be obtained. Thus, if the letters are $(a,\ b,\ c,\ d)$, and the operations are (as before) performed upon a^4, the entire series of columns is of the form (with successive columns laid out in alphabetical order):

$$\begin{array}{l|l|l|l|l|l|l|l|l|l|l}
a^4 & a^3b & a^3c & a^3d & a^2bd & a^2bcd & a^2c^2d & ab^2d^2 & abcd^2 & abd^3 & \cdots \\
 & a^2b^2 & a^2bc & a^2c^2 & a^2b^2c & a^2bd^2 & a^2cd^2 & a^2c^2d^2 & b^3cd & b^2cd^2 & d^4 \\
 & & ab^3 & a^2b^3 & ab^3d & ab^2cd & ab^2c^2 & b^4 & bcd^3 & c^2d^3 & \\
 & & & & b^4 & ab^3c & b^3c^2 & b^3d^2 & b^2d^3 & &
\end{array}$$

Nothing can be more convenient than the process when the entire series of columns is required; but it is very desirable to have a process for the formation of any column apart from the others; and the object of the present memoir is to investigate a rule for the purpose. But as showing how this rule is to be demonstrated, by very similar principles, I will commence with Arbogast's rule, applied as above, depends upon

If we take any combination b^2d and operate backwards on the last letter (viz. by changing it into its immediate antecessor in the alphabet), we obtain b^2c, which is a term in the next preceding column; b^2d is therefore obtainable from a term in the next preceding column, viz. b^2c, and the process is to operate on the last letter. If, instead of b^2d, the term is abc^2 (the last letter here entering as a power), the operation backwards on the last letter gives ab^2c, which is also a term of the preceding column; and it is to be noticed that the last but one letter b is here the immediate antecessor of the last letter c (and would have been so even if b had not entered into the given term abc^2, thus ac^2 operated on backwards would have given abc). Hence abc^2 is obtained by operating on a term in the next preceding column, viz. the term a^2bc, but in this case the operation is performed on the last but one letter. Every term is thus obtained from the next preceding column, viz. the terms are obtained by operating on the last letter, and (when the last but one letter is the immediate antecessor in the alphabet of the last letter) then also on the last but one letter, of each term of the next preceding column, and the correctness of the rule is thus demonstrated. It is to be observed that the terms are operated upon in order, the operation on the last but one letter (when it is operated on) being made immediately after that upon the last letter of the same term, and that the terms of a column are thus obtained in the proper alphabetical order.

I pass now to the above mentioned question of the formation of a single column by itself; it will be convenient, by way of illustration, to write down the columns

$$\begin{array}{lll}
a^2bd^2, & ac^2d^2, & c^2de^2, \\
ab^2d^2, & ab^2cd^2, & cd^2e^2, \\
a^2bce, & bc^2de, & bce^3, \\
a^2c^2e, & c^2de^2, & ad^4, \\
b^2ce^2, & ad^4, & bcd^3, \\
bc^2de, & bcd^3, & bd^4, \\
c^4e^2, & bd^4, & d^5;
\end{array}$$

which belong to the set (a, b, c, d, e), and are of the degree 5 and the weights 12 and 15 respectively.

Some definitions and explanations are required. I speak of the first and last letters of the set, simply as the first and the last letter: there is frequent occasion to speak of the last letter; and to avoid confusion, the last letter of a term will be spoken of as the *ultimate letter*; it is necessary also to consider the *penultimate, antepenultimate*, and *pro-antepenultimate letters* of the term. It will be convenient to distinguish between the ultimate letter and the *ultimate*, which may be either the ultimate letter or a power of such letter; and similarly for the penultimate, &c. Thus in the term bcd^2, the ultimate letter is d and the ultimate is d^2, the penultimate and the penultimate letter are each of them c; of course the ultimate, penultimate, &c. letters are always distinct from each other. We have also to consider the *pairs* of letters contained in a term; cd^2e contains the pairs (c, d), (c, e), (d, d), (d, e), and so in other cases; the letters of a pair are taken in the natural order. A pair not containing either the first letter or the last letter is *expansible*; thus the set being as before (a, b, c, d, e), the pairs (b, c), (d, d) are expansible: they are *expanded* by retreating the prior and advancing the posterior letter each one step; thus the just mentioned pairs (b, c), (d, d) are expanded into (a, d) and (c, e) respectively.

A pair composed of two distinct non-contiguous letters is *contractible*; it is contracted by advancing the prior and retreating the posterior letter each one step: thus (a, d), (c, e) are contractible pairs, and they give by contraction the pairs (b, c), (d, d) respectively: the processes of expansion and contraction are obviously converse to each other.

The expression *the last expansible pair* of a term hardly requires explanation; (d, d) is the last expansible pair of the term bcd^2, or of the term $b^2d^2e^2$, (c, d) the last expansible pair of the term c^2de (the set being always (a, b, c, d, e)), and so in all other cases. The expression *the last expansion*, in regard to any term, means the expansion of the last expansible pair of such term.

The expansion or contraction of any pair of a term leaves unaltered as well the weight as the degree, the resulting term belongs therefore to the same column as the original one. But the effect of an expansion is to diminish, and that of a contraction to increase the alphabetical rank, or *rank in the column*, the ranks being reckoned as the first or lowest, second, third, and so on, up to the last or highest rank. In particular, by performing upon any term the last expansion, we diminish the rank in the column, and by a succession of last expansions we bring the term up to the head of the column. Such term is not susceptible of any further expansion; it may therefore contain the first and last letters or either of them, and it may also contain a single intermediate letter in the first power only; thus the first and last letters being a, e, the head term of a column is of the form $a^m e^l$ or $a^m ce^l$, c being some intermediate letter, and the powers a^m, e^l being both or either of them omittable. In like manner, by a succession of contractions of any term we obtain the term at the foot of the column; such term is not susceptible of any further contraction, and it must therefore be composed either of a single letter or of two contiguous letters, that is it must be of the form c^m, or of the form $c^m d^l$, where c, d are any two contiguous letters, not excluding the case where the single letter or each or either of the contiguous letters is a first or a last letter.

It is to be observed that the passage up from any term to the term at the head of the column (or *top term*) by means of a succession of last expansions, is a perfectly unique one; but as no selection has been made of a like unique process of contraction, this is not the case with respect to the passage down from any term to the term at the foot of the column (or *bottom term*) by a succession of contractions.

Every term gives by the last expansion a term above it; it can therefore be obtained from such term above it by means of a contraction. But the contraction of the upper term is by hypothesis such that, performing upon the contracted term the last expansion, we obtain the upper term; a contraction, which, performed on any term, gives a lower term which by performing upon it the last expansion reproduces the first mentioned or upper term, may be called a *reversible contraction*. And it is clear that if we perform on the top term all the reversible contractions, and on each of the resulting terms all the reversible contractions, and so on as long as the process is possible, we obtain without repetition all the terms of the column. The column is in fact similar to a genealogical tree in the male line, each lower term issuing from a single upper term, while each upper term generates a lower term or terms, or does not generate any such term, and the top term being the common origin of the entire series. It may be added that when the order of the reversible contractions of the same term is duly fixed, the alphabetical arrangement of the terms in the column agrees with the order as of primogeniture (an elder son and his issue male preceding all the younger sons and their respective issue male) in the genealogical tree.

It only remains then to inquire under what conditions a contraction is reversible. Now as regards any term, in order that a contraction performed on it may be reversible, it is necessary and sufficient that the pair produced by the contraction should be the last expansible pair of the contracted term. There are several cases to be considered. First, if the contraction affects the ultimate and penultimate letters of the term: this implies that the ultimate and penultimate letters are not contiguous. Let the term terminate in $e^m h$, the contracted term will terminate in $e^{m-1} fg$, and (f, g), the pair produced by the contraction, is the last expansible pair of the contracted term; the contraction is in this case reversible. If, however, the term terminate in $e^m h^p$ $(p > 1)$, the contracted term will terminate in $e^{m-1} fgh^{p-1}$, and the last expansible pair is not as before (f, g), but it is (according as $p = 2$ or $p > 2$) (g, h) or (h, h): unless indeed h is the last letter, in which case (f, g) remains the last expansible pair of the contracted term. In the example e^m has been written, but the case $m = 1$ is not excluded; moreover, the penultimate letter e is removed three steps from the ultimate letter h, but the result would have been the same if instead of e we had had any preceding letter, or had had the letter f; by hypothesis it is not g, the letter contiguous to the ultimate letter h. The conclusion is that a contraction on the ultimate and penultimate letters (these being non-contiguous) is reversible if the ultimate is a simple letter, or if, being a power, it is a power of the last letter.

Next let the contraction affect the ultimate and antepenultimate letters. The two letters are here separated by the penultimate letter, and are therefore not contiguous. Suppose that the termination is $f^l g^m k$ (f and g contiguous), the contracted term terminates in $f^{l-1} g g^m j$, and (g, j) the pair arising from the contraction is the last expansible pair of the contracted term; the contraction is therefore reversible. In the example f^l has been written, but the case $l = 1$ is not excluded; moreover the ultimate letter k is taken non-contiguous to the penultimate letter g; but if the ultimate letter had been the contiguous letter h, the only difference is that the pair would be (g, g), and the conclusion is not altered. But suppose the termination of the term is $e^l g^m k$ (e, g non-contiguous), then the contracted term terminates in $e^{l-1} f g^m j$, where the pair arising from the contraction is (f, j), but the last expansible pair is (g, j); the contraction therefore is not reversible. The case $l = 1$ is not excluded; nor is it necessary that the ultimate letter should be non-contiguous to the penultimate; if the ultimate letter had been h, the pair arising from the contraction would have been (f, g), but the last expansible pair (g, g), and the contraction is still not reversible. In each of the cases considered the ultimate has been a simple letter; if in the first case the ultimate had been k^p ($p > 1$), then the contracted pair would terminate in jk^p, and (according as $p = 2$ or $p > 2$) the last expansible pair would be (j, k) or (k, k), which is not the pair (g, j) produced by the contraction; the contraction is therefore not reversible, unless indeed k is the last letter, in which case it continues reversible. In the second case the contraction, which is not reversible when the ultimate is the simple letter k, remains not reversible when the ultimate is a power of such letter. The conclusion is that a contraction on the ultimate and antepenultimate letters is reversible, if the penultimate and antepenultimate letters are contiguous, and the ultimate is a simple letter, or if, being a power, it is a power of the last letter.

A contraction on the ultimate and pro-antepenultimate letters, or on the ultimate letter and any letter preceding the pro-antepenultimate letter, is never reversible. To show this, it will be sufficient to consider the case where the term terminates in $efgh$, the contracted term terminates in $ff'gg$, the pair arising from the contraction being (f, g), and the last expansible pair being (g, g); and *à fortiori*, if the letters or any of them occur as powers, or are non-contiguous.

Consider, next, a contraction on the penultimate and antepenultimate letters; this assumes that these letters are non-contiguous. Such a contraction may be reversible if only the ultimate is the last letter or a power thereof; and the condition is then similar to that in the case of the ultimate and penultimate letters; only as the penultimate cannot be a power of the last letter, it must be a simple letter. And the conditions in order that the contraction may be reversible then are that the ultimate is the last letter or a power thereof, and the penultimate a simple letter.

The next case is that of a contraction on the penultimate and pro-antepenultimate letters. Such contraction may be reversible if the ultimate is the last letter or a power thereof; and the condition is then similar to that for the case of the ultimate and antepenultimate letters; only as the penultimate cannot be a power of the last letter, it must be a simple letter. The conditions, in order that the contraction may be reversible, are that the ultimate may be the last letter or a power thereof, the penultimate a simple letter, and the antepenultimate and pro-antepenultimate letters contiguous.

A contraction on the penultimate letter and on any letter preceding the pro-antepenultimate letter, or upon any two letters preceding the penultimate letter, is never reversible. If the ultimate, penultimate, antepenultimate and pro-antepenultimate letters are denoted by U, P, A, P' respectively, then by what precedes, the following contractions, viz. UP, UA, PA, PP', may be reversible, and the conditions for them are shown in the annexed Table. It is to be noticed that the conditions for UA, PA are mutually exclusive, and consequently that the number of reversible contractions to be performed upon any term is at most 3. The Table is

	A	P'
U	U, P, non-contiguous letters. The ultimate a simple letter, or a power of the last letter.	P, A, contiguous letters. The ultimate a simple letter, or a power of the last letter.
P	P, A, non-contiguous letters. Penultimate a simple letter. Ultimate the last letter, or a power thereof.	A, P', contiguous letters. Penultimate a simple letter. Ultimate the last letter, or a power thereof.

The contractions are to be applied in the order UP, UA, PA, PP', but all the contracted terms originating in a prior contraction of a given term are to be exhausted before proceeding to a posterior contraction of the same term. As an example of the process, the set being (a, b, c, d, e), I will take the column

$$\begin{array}{l} abe^3, \\ acde^2, \\ ad^3e, \\ b^2de^2, \\ bc^2e^2, \\ bcd^2e, \\ bd^4, \\ c^2d^2e, \\ c^4e. \end{array}$$

The top term abe^3 is given. The contractions applicable to it are UP, UA. And UP gives $acde^2$. The only contraction applicable to this is UA, giving ad^3e. And there is not any contraction applicable to ad^3e. We revert therefore to UA on abe^3, this gives b^2de^2. The only applicable contraction is PA, giving bc^2e^2. The contractions applicable to this are UP, UA. And UP gives bcd^2e. The only contraction applicable to this is UA, giving bd^4, and there is not any contraction applicable to bd^4. We revert therefore to UA on bc^2e^2, this gives c^2d^2e, and the only contraction applicable to this is UA, giving c^4e. There is not any contraction applicable to this, and the process is therefore complete. It may be remarked that the example presents no instance of the contraction PP'; in fact the only terms having a pro-antepenultimate are $acde^2$, in which A, P' are not contiguous letters, and bcd^2e, in which the penultimate is not a simple letter, so that the contraction PP' is in each case excluded. It must be confessed that a considerable amount of practice would be required before the process could be readily made use of.

265.

Addition to the Memoir on an Extension of Arbogast's Method of Derivations.

[Now first published, 1891.]

The process explained in the foregoing Memoir is not easily workable, and I have in fact never used it. I have devised a new process theoretically less complete, and of a somewhat mixed character consisting, as it does, in the analytical reduction of the problem to the same problem for a smaller number of letters. I have however found it very convenient for obtaining the literal terms in the theory of the sextic, viz. where we have the seven letters (a, b, c, d, e, f, g); and I propose to explain the process by applying it to the determination of the terms of the discriminant, degree $= 6$, weight $= 18$.

Here treating a, b, c, d, e, f, g to denote the indices of these letters respectively in a term such as $a^a b^b c^c \cdots$ (that is, writing for convenience $a, b, c, \ldots$ instead of $\alpha, \beta, \gamma, \ldots$) we have

$$a + b + c + d + e + f + g = 6,$$

and thence

$$b + 2c + 3d + 4e + 5f + 6g = 18,$$

$$6a + 5b + 4c + 3d + 2e + f = 18.$$

I separate off from the others the first three letters a, b, c. The equation gives $a = 3$ at most. And then

$$\begin{array}{llll}
\text{if} \quad a = 3, & \text{then } \ 5b + 4c + 3d + 2e + f = 0, & & b = 0,\ c = 0; \\
\text{if} \quad a = 2, & 5b + 4c + 3d + 2e + f = 6, & & b = 1 \text{ at most}; \\
& \text{if } b = 1,\ 4c + 3d + 2e + f = 1, & & c = 0; \\
& \text{if } b = 0,\ 4c + 3d + 2e + f = 6, & & c = 1 \text{ at most.}
\end{array}$$

if $a=1$, $5b+4c+3d+2e+f=12$, $b=2$ at most;
if $b=2$, $4c+3d+2e+f=2$, $c=0$;
if $b=1$, $4c+3d+2e+f=7$, $c=1$ at most;
if $b=0$, $4c+3d+2e+f=12$, $c=3$ at most;

if $a=0$, $5b+4c+3d+2e+f=18$,
if $b=3$, $4c+3d+2e+f=3$, $c=0$;
if $b=2$, $4c+3d+2e+f=8$, $c=2$ at most;
if $b=1$, $4c+3d+2e+f=13$, $c=3$ at most;
if $b=0$, $4c+3d+2e+f=18$, $c=4$ at most.

Hence considering the several terms as arranged in alphabetical order and writing down only the factors in a, b, c, we obtain col. 1 of the following diagram;

Col. 1.	Col. 2.	Col. 3.
a^3	$(d^3)^9$	1
a^2b	$(d^3)^8$	1
a^2c	$(d^3)^7$	2
a^2	$(d^4)^6$	5
ab^2	$(d^3)^7$	2
abc	$(d^3)^6$	3
ab	$(d^4)^5$	4
ac^3	$(d^2)^6$	1
ac^2	$(d^3)^5$	3
ac	$(d^4)^4$	4
a	$(d^5)^3$	3
b^3	$(d^3)^6$	3
b^2c^2	$(d^2)^6$	1
b^2c	$(d^3)^5$	3
b^2	$(d^4)^4$	4
bc^3	$(d^2)^5$	1
bc^2	$(d^3)^4$	3
bc	$(d^4)^3$	3
b	$(d^5)^2$	2
c^4	$(d^2)^4$	2
c^3	$(d^3)^3$	3
c^2	$(d^4)^2$	2
c	$(d^5)^1$	1
1	$(d^6)^0$	1
		58

We then form col. 2 by annexing to each term a term $(d^\theta)^\varphi$ which denotes the derivative φ of d^θ; the value of θ being such that the whole term may be of the proper degree 6, and the value of φ being such that the whole term may be of the proper weight 18. Thus

$$a^3(d^3)^9, \quad \text{degree is } 3+3, \ =6\ ; \quad \text{weight is } 0+9+9, \ =18,$$
$$a^2bc\,(d^3)^7, \quad \text{,,}\ 1+2+3, \ =6\ ; \quad \text{,, ,,}\ 0+2+9+7, \ =18,$$

viz. d^3 being of weight 9, then $(d^3)^9$ is of weight $9+9$, and $(d^3)^7$ of weight $9+7$.

The derivatives of any power of d are given by those of the indefinite power d^m, which, omitting therein the several powers of d, may be tabulated thus. I remark that the table is carried up to the column 21 in order that it may be applicable to the calculation of the literal terms of the degree 15 and weight 45 which belong to the highest invariant of the sextic.

Subsidiary Table of the Derivatives of d^m.

	1	1	2	3	4	5	7	8	10	12	14	16	19	21	24	27	30	33	37	40	44	48	
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	
1	1	e	e^2	e^3	e^4	e^5	e^6	e^7	e^8	e^9	e^{10}	e^{11}	e^{12}	e^{13}	e^{14}	e^{15}	e^{16}	e^{17}	e^{18}	e^{19}	e^{20}	e^{21}	1
2			f	ef	e^2f	e^3f	e^4f	e^5f	e^6f	e^7f	e^8f	e^9f	$e^{10}f$	$e^{11}f$	$e^{12}f$	$e^{13}f$	$e^{14}f$	$e^{15}f$	$e^{16}f$	$e^{17}f$	$e^{18}f$	$e^{19}f$	2
3				g	f^2	ef^2	e^2f^2	e^3f^2	e^4f^2	e^5f^2	e^6f^2	e^7f^2	e^8f^2	e^9f^2	$e^{10}f^2$	$e^{11}f^2$	$e^{12}f^2$	$e^{13}f^2$	$e^{14}f^2$	$e^{15}f^2$	$e^{16}f^2$	$e^{17}f^2$	3
4					eg	e^2g	f^3	ef^3	e^2f^3	e^3f^3	e^4f^3	e^5f^3	e^6f^3	e^7f^3	e^8f^3	e^9f^3	$e^{10}f^3$	$e^{11}f^3$	$e^{12}f^3$	$e^{13}f^3$	$e^{14}f^3$	$e^{15}f^3$	4
5						fg	e^3g	e^4g	f^4	ef^4	e^2f^4	e^3f^4	e^4f^4	e^5f^4	e^6f^4	e^7f^4	e^8f^4	e^9f^4	$e^{10}f^4$	$e^{11}f^4$	$e^{12}f^4$	$e^{13}f^4$	5
6							efg	e^2fg	e^5g	e^6g	f^5	ef^5	e^2f^5	e^3f^5	e^4f^5	e^5f^5	e^6f^5	e^7f^5	e^8f^5	e^9f^5	$e^{10}f^5$	$e^{11}f^5$	6
7							g^2	f^2g	e^3fg	e^4fg	e^7g	e^8g	f^6	ef^6	e^2f^6	e^3f^6	e^4f^6	e^5f^6	e^6f^6	e^7f^6	e^8f^6	e^9f^6	7
8								eg^2	ef^2g	e^2f^2g	e^5fg	e^6fg	e^9g	$e^{10}g$	f^7	ef^7	e^2f^7	e^3f^7	e^4f^7	e^5f^7	e^6f^7	e^7f^7	8
9									e^2g^2	f^3g	e^3f^2g	e^4f^2g	e^7fg	e^8fg	$e^{11}g$	$e^{12}g$	f^8	ef^8	e^2f^8	e^3f^8	e^4f^8	e^5f^8	9
10									fg^2	e^3g^2	ef^3g	e^2f^3g	e^5f^2g	e^6f^2g	e^9fg	$e^{10}fg$	$e^{13}g$	$e^{14}g$	f^9	ef^9	e^2f^9	e^3f^9	10
11										efg^2	e^4g^2	f^4g	e^3f^3g	e^4f^3g	e^7f^2g	e^8f^2g	$e^{11}fg$	$e^{12}fg$	$e^{15}g$	$e^{16}g$	f^{10}	ef^{10}	11
12										g^3	e^2fg^2	e^5g^2	ef^4g	e^2f^4g	e^5f^3g	e^6f^3g	e^9f^2g	$e^{10}f^2g$	$e^{13}fg$	$e^{14}fg$	$e^{17}g$	$e^{18}g$	12
13											f^2g^2	e^3fg^2	e^6g^2	f^5g	e^3f^4g	e^4f^4g	e^7f^3g	e^8f^3g	$e^{11}f^2g$	$e^{12}f^2g$	$e^{15}fg$	$e^{16}fg$	13
14											eg^3	ef^2g^2	e^4fg^2	e^7g^2	ef^5g	e^2f^5g	e^5f^4g	e^6f^4g	e^9f^3g	$e^{10}f^3g$	$e^{13}f^2g$	$e^{14}f^2g$	14
15												e^2g^3	$e^2f^2g^2$	e^5fg^2	e^8g^2	f^6g	e^3f^5g	e^4f^5g	e^7f^4g	e^8f^4g	$e^{11}f^3g$	$e^{12}f^3g$	15
16												fg^3	f^3g^2	$e^3f^2g^2$	e^6fg^2	e^9g^2	ef^6g	e^2f^6g	e^5f^5g	e^6f^5g	e^9f^4g	$e^{10}f^4g$	16
17													e^3g^3	ef^3g^2	$e^4f^2g^2$	e^7fg^2	$e^{10}g^2$	f^7g	e^3f^6g	e^4f^6g	e^7f^5g	e^8f^5g	17
18													efg^3	e^4g^3	$e^2f^3g^2$	$e^5f^2g^2$	e^8fg^2	$e^{11}g^2$	ef^7g	e^2f^7g	e^5f^6g	e^6f^6g	18
19													g^4	e^2fg^3	f^4g^2	$e^3f^3g^2$	$e^6f^2g^2$	e^9fg^2	$e^{12}g^2$	f^8g	e^3f^7g	e^4f^7g	19
20														f^2g^3	e^5g^3	ef^4g^2	$e^4f^3g^2$	$e^7f^2g^2$	$e^{10}fg^2$	$e^{13}g^2$	ef^8g	e^2f^8g	20
21														eg^4	e^3fg^3	e^6g^3	$e^2f^4g^2$	$e^5f^3g^2$	$e^8f^2g^2$	$e^{11}fg^2$	$e^{14}g^2$	f^9g	21
22															ef^2g^3	e^4fg^3	f^5g^2	$e^3f^4g^2$	$e^6f^3g^2$	$e^9f^2g^2$	$e^{12}fg^2$	$e^{15}g^2$	22
23															e^2g^4	$e^2f^2g^3$	e^7g^3	ef^5g^2	$e^4f^4g^2$	$e^7f^3g^2$	$e^{10}f^2g^2$	$e^{13}fg^2$	23
24															fg^4	f^3g^3	e^5fg^3	e^8g^3	$e^2f^5g^2$	$e^5f^4g^2$	$e^8f^3g^2$	$e^{11}f^2g^2$	24
25																e^3g^4	$e^3f^2g^3$	e^6fg^3	f^6g^2	$e^3f^5g^2$	$e^6f^4g^2$	$e^9f^3g^2$	25
26																efg^4	ef^3g^3	$e^4f^2g^3$	e^9g^3	ef^6g^2	$e^4f^5g^2$	$e^7f^4g^2$	26
27																g^5	e^4g^4	$e^2f^3g^3$	e^7fg^3	$e^{10}g^3$	$e^2f^6g^2$	$e^5f^5g^2$	27
28																	e^2fg^4	f^4g^3	$e^5f^2g^3$	e^8fg^3	f^7g^2	$e^3f^6g^2$	28
29																	f^2g^4	e^5g^4	$e^3f^3g^3$	$e^6f^2g^3$	$e^{11}g^3$	ef^7g^2	29
30																	eg^5	e^3fg^4	ef^4g^3	$e^4f^3g^3$	e^9fg^3	$e^{12}g^3$	30
31																		ef^2g^4	e^6g^4	$e^2f^4g^3$	$e^7f^2g^3$	$e^{10}fg^3$	31
32																		e^2g^5	e^4fg^4	f^5g^3	$e^5f^3g^3$	$e^8f^2g^3$	32
33																		fg^5	$e^2f^2g^4$	e^7g^4	$e^3f^4g^3$	$e^6f^3g^3$	33
34																			f^3g^4	e^5fg^4	ef^5g^3	$e^4f^4g^3$	34
35																			e^3g^5	$e^3f^2g^4$	e^8g^4	$e^2f^5g^3$	35
36																			efg^5	ef^3g^4	e^6fg^4	f^6g^3	36
37																			g^6	e^4g^5	$e^4f^2g^4$	e^9g^4	37
38																				e^2fg^5	$e^2f^3g^4$	e^7fg^4	38
39																				f^2g^5	f^4g^4	$e^5f^2g^4$	39
40																				eg^6	e^5g^5	$e^3f^3g^4$	40
41																					e^3fg^5	ef^4g^4	41
42																					ef^2g^5	e^6g^5	42
43																					e^2g^6	e^4fg^5	43
44																					fg^6	$e^2f^2g^5$	44
45																						f^3g^5	45
46																						e^3g^6	46
47																						efg^6	47
48																						g^7	48

This means for instance that the derivatives of d^2 are

0	1	2	3	4	5	6
d^2	de	d^2g d^2f def	deg df^2 de^2g	dfg dg^2	$\dots$	$\dots$

those of d^3 are

0	1	2	3	4	5	6	7	8	9
d^3	d^2e	d^2f	d^2g	deg	df^2	dg^2	$\dots$	$\dots$	$\dots$

and so on; viz. to form any column we affix to the terms of the corresponding column of the subsidiary table the proper power of d, so that the degree in all the letters may be $2, 3,$ &c. as the case may be, using from each column of the subsidiary table only the terms which are not of too high a degree; for instance for col. 4 of the derivatives of d^2, only the terms eg, f^2, e^2f of the complete column eg, f^2, e^2f, e^4.

It is to be observed that these tables of the derivatives of d^2, d^3, &c. do not need to be actually formed; any column which is wanted can be written down at once, *currente calamo*, from the column of the subsidiary table. And if we require only the number of terms, then these numbers are at once taken out from the Subsidiary Table. Thus for the numerical column of the diagram, the Subsidiary Table, col. 9, contains but 1 term g^3 of degree 3, and thus the number set opposite to $(d^3)^9$ is 1; so col. 8 contains but 1 term fg^2 of the degree 3, and so the number opposite to $(d^3)^8$ is 1; col. 7 contains 2 terms eg^2, f^2g of degree 3, and so the number opposite to $(d^3)^7$ is 2; and so on: we have in this way the numbers $1, 1, 2, 5, 2, 3, 4,$ &c. the partial sums of which are $1, 1, 7, 9, 11, 3, 8, 9, 9$ giving the total sum 58: viz. this is the number of the terms degree 6, weight 18, which can be formed with the seven letters (a, b, c, d, e, f, g).

When the literal terms are required, it is proper in the first instance as a safeguard against accidental omissions in copying, to take the numbers in this manner; and we can then take out the terms themselves, viz. these are

$$\begin{array}{l} a^3b\,d^3fg^2, \quad a^3c\,d^2efg, \quad \dots, \\ a^2b^2c\,d^2e^2f, \quad \dots, \\ \dots \end{array}$$

and so on, the whole number being $= 58$ as just mentioned.

266.

On the Equation for the Product of the Differences of all but One of the Roots of a Given Equation.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. for the year 1861, pp. 45–59. Received November 30, 1860,—Read January 10, 1861.]

It is easy to see that for an equation of the order n, the product of the differences of all but one of the roots will be determined by an equation of the order n, the coefficients of which are alternately rational functions of the coefficients of the original equation, and rational functions multiplied by the square root of the discriminant. In fact, if the equation be $\varphi v = (a, \dots \mid v, 1)^n = a\,(v-\alpha)(v-\beta)\cdots$, then putting for the moment $a=1$, and disregarding numerical factors, $\sqrt{D}$, the square root of the discriminant, is equal to the product of the differences of the roots, and $\varphi'\alpha$ is equal to $(\alpha-\beta)(\alpha-\gamma)\cdots$, consequently the product of the differences of the roots, all but α, is equal to $\sqrt{D} \div \varphi'\alpha$, and the expression $\dfrac{\sqrt{D}}{\varphi'\alpha}$ is the root of an equation of the order n, the coefficients of which are rational functions of the coefficients of the original equation. I propose in the present memoir to determine the equation in question for equations of the orders three, four, and five: the process employed is similar to that in my memoir "On the Equation of Differences for an Equation of any Order, and in particular for Equations of the Orders Two, Three, Four, and Five," *Phil. Trans.*, vol. CL. (1860), [262], viz. the last coefficient of the given equation is put equal to zero, so that the given equation breaks up into $v=0$ and into an equation of the order $n-1$ called the reduced equation; and this being so, the required equation breaks up into an equation of the order $n-1$ (which however is not, as for the equation of differences, that which corresponds to the reduced equation) and into a linear equation; the equation of the order $n-1$ is calculated by the method of symmetric functions; and combining it with the linear equation, which is known, we have the required equation, except as regards the terms involving the last coefficient, which terms are found by the consideration that the coefficients of the required equation are seminvariants. The solution leads immediately to that of a more general question; for if the product of the differences of all the roots except α, of the given equation

$$\varphi v = (* \mid v, 1)^n = a\,(v-\alpha)(v-\beta)\cdots = 0$$

(which product is a function of the degree $n-2$ in regard to each of the roots $\beta, \gamma, \delta, \ldots$), is multiplied by $(x-\alpha y)^{n-2}$, the function so obtained will be the root of an equation of the order n, the coefficients of which are covariants of the quantic $(* \mid x, y)^n$, and these coefficients can be at once obtained by writing, in the place of the seminvariants of the former result, the covariants to which they respectively belong. In the case of the quintic equation, one of these covariants is, in regard to the coefficients, of the degree 6, which exceeds the limit of the tabulated covariants, [the covariants are all tabulated, 141 and 143], the covariant in question has therefore to be now first calculated. The covariant equations for the cubic and the quartic might be deduced from the formulæ Nos. 119 and 142 of my Fifth memoir on Quantics, *Phil. Trans.*, vol. CXLVIII. (1858), [156]; they are in fact the bases of the methods which are there given for the solution of the cubic and the quartic equations respectively; and it was in this way that I was led to consider the problem which is here treated of.

1. The notation $\zeta^2(\alpha, \beta, \gamma, \ldots)$ is used (after Professor Sylvester) to denote the product of the squared differences of $(\alpha, \beta, \gamma, \ldots)$, and the notation $\zeta^4(\alpha, \beta, \gamma, \ldots)$ to denote the product of the differences taken in a determinate order, viz.

$$\begin{aligned}\zeta^4(\alpha, \beta, \gamma, \delta, \ldots) = {} & (\alpha-\beta)(\alpha-\gamma)(\alpha-\delta)\cdots \\ & (\beta-\gamma)(\beta-\delta)\cdots \\ & (\gamma-\delta)\cdots\end{aligned}$$

2. The product of the differences of the roots of an equation depends, as already noticed, on the square root of the discriminant; and in order to fix the numerical factors and signs, it will be convenient, in regard to the equations

$$\begin{aligned}&(a, b, c \mid v, 1)^2 = 0,\\ &(a, b, c, d \mid v, 1)^3 = 0,\\ &(a, b, c, d, e \mid v, 1)^4 = 0,\\ &(a, b, c, d, e, f \mid v, 1)^5 = 0,\end{aligned}$$

to write as follows:

$$\zeta^4(\alpha,\beta) = \tfrac{1}{a}\sqrt{-(4ac-b^2)} = \tfrac{1}{a}\sqrt{-\square},$$
$$\zeta^4(\alpha,\beta,\gamma) = \tfrac{1}{a^2}\sqrt{-(27\,a^2d^2-\cdots)} = \tfrac{1}{a^2}\sqrt{-\square},$$
$$\zeta^4(\alpha,\beta,\gamma,\delta) = -\tfrac{1}{a^3}\sqrt{256\,a^3e^3-27\,a^2d^4+\cdots} = -\tfrac{1}{a^3}\sqrt{\square},$$
$$\zeta^4(\alpha,\beta,\gamma,\delta,\varepsilon) = -\tfrac{1}{a^4}\sqrt{3125\,a^4f^4+256\,a^4e^5+\cdots} = -\tfrac{1}{a^4}\sqrt{\square},$$

where it is to be observed, for example, that writing in the last equation $e=0$, and therefore $f=0$, we have $\zeta^4(\alpha,\beta,\gamma,\delta,0)=\frac{1}{a^4}\sqrt{256\,a^4e^5+\cdots}$, which agrees with the equation $\zeta^4(\alpha,\beta,\gamma,\delta,0) = \alpha\beta\gamma\delta\,\zeta^4(\alpha,\beta,\gamma,\delta) = \frac{e}{a}\,\zeta^4(\alpha,\beta,\gamma,\delta)$, if for $\zeta^4(\alpha,\beta,\gamma,\delta)$ we substitute the value given by the last equation but one.

For the cubic equation $(a,b,c,d \mid v,1)^3=0$;

3. We have to find the equation for $\theta=\zeta^2(\alpha,\beta)=(\alpha-\beta)^2$; the roots are

$$\theta_1=\beta-\gamma,\quad \theta_2=\gamma-\alpha,\quad \theta_3=\alpha-\beta.$$

To apply the method above explained, write $\gamma=0$, and therefore also $d=0$; the roots thus become

$$\theta_1=\beta,\quad \theta_2=-\alpha,\quad \theta_3=\alpha-\beta,$$

and we have the quadric and linear equations

$$(\theta+\alpha)(\theta-\beta)=0, \qquad \theta-(\alpha-\beta)=0,$$

where (α,β) are the roots of the equation

$$(a,b,c \mid v,1)^2=0.$$

Hence, writing

$$\square = 4ac-b^2,$$

we have

$$\alpha-\beta = -\sqrt{-\square}\,/\,a,$$

and the two equations become

$$\theta^2-\theta\big[(\alpha+\beta)+(\alpha-\beta)\big]+(-\alpha\beta)-\alpha(\alpha-\beta)=0, \qquad \theta-(\alpha-\beta)=0,$$

or multiplying the two equations together,

$$\theta^3\,a^2+\theta^2\cdot 0+\theta\,(3ac-b^2)+c\sqrt{-\square}=0,$$